# Enhanced diffusion over a periodic trap by hydrodynamic coupling to an elastic mode – *Supplementary Information* –

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(Dated: October 14, 2025)

## I. SUPPLEMENTARY NOTE 1: DESCRIPTION OF THE MODEL

We consider an overdamped system made up of two degrees of freedom, the principal degree of freedom $q_1$ which is free to explore a large region of space, for example subjected to a periodic potential $\phi(q_1)$, and the second degree of freedom $q_2$, highly constrained and taken to be harmonically confined with potential energy $\frac{1}{2\kappa}q_2^2$, where we assume that $\kappa$ the spring compliance is small. The total energy of the system is thus given by the Hamiltonian

$$H(q_1, q_2) = \phi(q_1) + \frac{1}{2\kappa}q_2^2 \; . \tag{S1}$$

In thermal equilibrium, the steady state of the probability distribution of $q_1$ and $q_2$ is given by the Gibbs-Boltzmann distribution

$$P_{\rm eq}(q_1, q_2) = \frac{\exp\left[-\beta H(q_1, q_2)\right]}{Z} \; , \tag{S2}$$

where $Z$ is the normalizing partition function and $\beta = 1/(k_{\rm B}T)$ is the inverse thermal energy, where $T$ is the temperature and $k_{\rm B}$ is Boltzmann's constant. In equilibrium, the statistics of $q_1$ and $q_2$ are independent due to the additivity of the Hamiltonian $H$, *i.e.* $P_{eq}(q_1, q_2) = P_{1eq}(q_1)P_{2eq}(q_2)$. However, dynamically, they can be coupled; for instance this model can be seen as a toy model for a colloidal particle near an elastic compliant surface. As such, any single time observation in equilibrium of the measurable quantity $q_1$ will be independent of the effect of $q_2$ and hence will not show any effect due to the hydrodynamic coupling with $q_2$. This means that one would have to measure at least two time correlation functions; but, due to the rapid dynamics of $q_2$, one would have to resolve the dynamics on very short time scales to detect the hydrodynamic coupling.

The simplest form for the over-damped Langevin dynamics of this system is

$$\frac{{\rm d}q_1}{{\rm d}t} = -\mu_{11}\phi'(q_1) - \frac{\mu_{12}}{\kappa}q_2 + \eta_1(t) \tag{S3}$$

$$\frac{{\rm d}q_2}{{\rm d}t} = -\frac{\mu_{22}}{\kappa}q_2 - \mu_{12}\phi'(q_1) + \eta_2(t) \; , \tag{S4}$$

where the mobility coefficients $\mu_{ij}$ are taken for simplicity to be constant, and the noise correlation function is chosen to ensure that this Langevin system has a steady state distribution given by Eq. (S2), explicitly

$$\langle \eta_i(t)\eta_j(t)\rangle = 2k_{\rm B}T\delta(t-t')\mu_{ij} \; . \tag{S5}$$

From a mathematical standpoint, the constraints on the mobility matrix are that it is symmetric and has positive eigenvalues. However, from a physical point of view, in the absence of coupling, we must have

$$\mu_{11} > 0, \text{ and } \mu_{22} > 0 \; . \tag{S6}$$

The constraints on the eigenvalues of the mobility matrix then translate to

$$\det(\mu) = \mu_{11}\mu_{22} - \mu_{12}^2 > 0 \; . \tag{S7}$$

In an experimental context, it may be possible to directly observe the motion of the variable $q_1$, but not necessarily on very short time scales, while the direct observation of the second variable $q_2$ may be impossible for it cannot be

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resolved spatially nor temporally. If the variable $q_1$ is subject to a confining potential and a measurement is made of its equilibrium statistics, the variable $q_1$ has no effect whatsoever, and changing the value of $\kappa$ for instance will not affect the measurements. Observing the dynamics of the variable $q_1$ may yield information about the variable $q_2$ but if the time scale over which $q_2$ equilibrates is too short, again this information is censored by being averaged out.

Here, we propose a solution to demonstrate the effect of the variable $q_2$ by examining its effect on the late time diffusion constant of $q_1$ when it is subjected to a periodic potential. Given that we are examining diffusion at large times and over large distances, neither the temporal nor the spatial resolutions of the experiment are an issue. The effect of the variable $q_2$ is determined by its influence on how $q_1$ moves in the potential $\phi$ and how this influence integrates over time to change the diffusion constant of $q_1$. We will see that there is an intrinsic time scale associated with the dynamics of the elastic variable $q_2$, and we assume that this time scale is small and thus difficult to resolve. Measurements of $q_1$ made over time scales larger than $\tau_2$ will not see the effect of $q_2$ due to averaging. If the variable $q_1$ diffuses in a periodic potential and the typical time to cross the potential barriers (*i.e.* jump from minimum to minimum) is of the order of $\tau_2$, then the first passage time from minimum to minimum will be affected by the dynamics of $q_2$ and thus show up in the late time diffusion constant of $q_1$, yielding a mechanism for revealing the hydrodynamic coupling. Rather than trying to resolve short time scales directly, we are thus imposing a short distance, and this can be done with an optical trap.

## II. SUPPLEMENTARY NOTE 2: HEURISTIC ARGUMENT FOR THE CORRECTION TO THE LATE TIME DIFFUSION CONSTANT

To determine the effective Langevin dynamics for the variable $q_1$ in the limit where $\kappa$ is small, and so the variable $q_2$ is small, we consider the dynamics of the system without any noise. Using the subsequently derived effective dynamics for $q_1$ at zero temperature, we then add a white noise to the system in order to ensure that the resulting Langevin equation has the correct marginal equilibrium probability distribution

$$P_{\rm meq}(q_1) = \frac{\exp\left[-\beta\phi(q_1)\right]}{Z_-} \ , \tag{S8}$$

where $\beta = \frac{1}{k_{\rm B}T}$ is the inverse thermal energy, and where we use the notation (made dimensionless as compared to the main text)

$$Z_\pm = \int_0^\lambda {\rm d}q \exp[\pm\beta\phi(q)] \ , \tag{S9}$$

where $\lambda$ is the period of the potential. This is clearly not a rigorous procedure, and we will justify the result rigorously below using multiscale analysis. This method can also be applied to compute the correction to diffusion of a Brownian particle in a periodic potential due to inertia (the deviation from the strongly over-damped limit) [1, 2] and actually recovers first order correction (for completeness, we give a derivation later in this SM). It can also be applied to compute the correction to the diffusion constant of a Brownian particle carrying an electric dipole in a periodic electric field; and again, the simple analysis agrees with a more rigorous one using stochastic analysis [3].

We proceed by formally solving Eq. (S4) at $T = 0$ (that is to say, without noise) to give

$$q_2 = -\frac{\kappa\mu_{12}}{\mu_{22}}\left(1 + \frac{\kappa}{\mu_{22}}\frac{\rm d}{{\rm d}t}\right)^{-1}\phi'(q_1) \ , \tag{S10}$$

where $\tau_2 = \frac{\kappa}{\mu_{22}}$ is a short time scale due to the fact that $\kappa$ is small. Carrying out an operator expansion for small $\tau_2$, we find

$$q_2 \simeq -\frac{\kappa\mu_{12}}{\mu_{22}}\left[\phi'(q_1) - \frac{\kappa}{\mu_{22}}\phi''(q_1)\frac{{\rm d}q_1}{{\rm d}t} + O(\tau_2^2)\right]. \tag{S11}$$

Now, substituting this in Eq. (S3) gives

$$\frac{{\rm d}q_1}{{\rm d}t} = -\left(\mu_{11} - \frac{\mu_{12}^2}{\mu_{22}}\right)\phi'\left(q_1(t)\right) - \frac{\kappa\mu_{12}^2}{\mu_{22}^2}\phi''(q_1(t))\frac{{\rm d}q_1}{{\rm d}t} \ , \tag{S12}$$

and from this

$$\frac{{\rm d}q_1}{{\rm d}t} = -\mu_{\rm e}(q_1)\phi'(q_1) \tag{S13}$$

where $\mu_\mathrm{e}(q_1)$ is the effective mobility

$$\mu_\mathrm{e}(q_1) = \frac{\mu_{11}\mu_{22} - \mu_{12}^2}{\mu_{22}\left(1 + \frac{\kappa\phi''(q_1)\mu_{12}^2}{\mu_{22}^2}\right)} \tag{S14}$$

In terms of the friction matrix $\gamma = \mu^{-1}$, the effective mobility can be written in a simpler form

$$\mu_\mathrm{e}(q_1) = \frac{1}{\gamma_{11}\left(1 + \frac{\kappa\phi''(q_1)\gamma_{12}^2}{\gamma_{11}^2}\right)} \ . \tag{S15}$$

Rather than discussing the noise necessary to render this into a Langevin equation, we note that to have the correct equilibrium distribution, the corresponding Fokker-Planck equation must read

$$\frac{\partial P_1(q_1,t)}{\partial t} = \frac{\partial}{\partial q_1}\mu_\mathrm{e}(q_1)\left[k_\mathrm{B}T\frac{\partial P_1(q_1,t)}{\partial q_1} + \phi'(q_1)P_1(q_1,t)\right] \ , \tag{S16}$$

and we see that the effective diffusion constant is given by $D_\mathrm{e}(q_1) = k_\mathrm{B}T\mu(q_1)$. Note that in doing this, we have added multiplicative noise and a temperature dependent drift if we use the Ito convention of stochastic calculus; however we avoid this discussion by working directly with the Fokker-Planck equation. Notice that when $T = 0$, Eq. (S16) is the transport equation for Eq. (S13).

Now, if the potential is periodic with period $\lambda$, we can use the well known Lifson-Jackson formula for the late time effective diffusion constant for Langevin processes in a potential and with varying diffusivity, both of period $\lambda$:

$$D^* = \frac{\lambda^2}{\int_0^\lambda \mathrm{d}q \frac{\exp[\beta\phi(q)]}{D_\mathrm{e}(q)} \int_0^\lambda \mathrm{d}q \exp[-\beta\phi(q)]} \ . \tag{S17}$$

In this case, we can easily show by integration by parts that

$$D^*(\kappa) = \frac{k_\mathrm{B}T\lambda^2}{\gamma_{11}\int_0^\lambda \mathrm{d}q \exp[-\beta\phi(q)] \int_0^\lambda \mathrm{d}q \exp[\beta\phi(q)]\left(1 - \frac{\kappa\gamma_{12}^2\phi'^2(q)}{k_\mathrm{B}T\gamma_{11}^2}\right)} \ . \tag{S18}$$

We thus see that in the regime where $\kappa$ is small, the effect of a finite value of $\kappa$ is to increase the long time diffusivity of the particle. We can also write $\phi(q) = \Delta U\psi\left(\frac{q}{\lambda}\right)$, where $\Delta U$ is an adjustable energy scale (for instance, it could be related to the laser intensity and in which case $\lambda$ would be a wavelength). This allows us to write

$$D^*(\kappa) = \frac{k_\mathrm{B}T}{\gamma_{11}\int_0^1 \mathrm{d}p \exp[-\beta\Delta U\psi(p)] \int_0^1 \mathrm{d}p \exp\left[\beta\Delta U\psi(p)\right]\left(1 - \frac{\kappa\Delta U^2\psi'^2(p)\gamma_{12}^2}{k_\mathrm{B}T\lambda^2\gamma_{11}^2}\right)} \ . \tag{S19}$$

Therefore, even if $\kappa$ is very small, increasing the value of $\Delta U$ or decreasing the value of $L$ can be used to make the change in the long time diffusion constant measurable.

In the absence of any compliance, $\kappa = 0$, and no potential we note that

$$D^*(0) = \frac{k_\mathrm{B}T}{\gamma_{11}} \ , \tag{S20}$$

this corresponds to the modification of the diffusivity of $q_1$ when the *surface* $q_1$ is rigid but still exerts a hydrodynamic interaction.

From a physical standpoint, the fluctuations of $q_2$ can only influence the dynamics of $q_1$ on the time scale $\tau_2$ before they are averaged out. However, if the typical time that $q_1$ takes to climb over the potential barrier to the next period is of order $\tau_2$, then $q_2$ can effect the long time diffusion of $q_1$. In this way, fast and small degrees of freedom coupled to a slower degree of freedom can be detected even at low temporal resolution, provided one has sufficiently good statistics. By changing variables, we obtain

$$\frac{D^*(\kappa)}{D^*(0)} = \left(1 - \frac{\int_0^1 \mathrm{d}p \exp[\beta\Delta U\psi(p)]\frac{\kappa\Delta U^2\psi'^2(p)\gamma_{12}^2}{k_\mathrm{B}T\lambda^2\gamma_{11}^2}}{\int_0^1 \mathrm{d}p \exp[\beta\Delta U\psi(p)]}\right)^{(-1)} \ . \tag{S21}$$

Now, setting

$$v = \beta \Delta U \tag{S22}$$

gives

$$\frac{D^*(\kappa)}{D^*(0)} = \left(1 - \frac{\kappa k_\mathrm{B} T v^2 \gamma_{12}^2}{\lambda^2 \gamma_{11}^2} \frac{\int_0^1 \mathrm{d}p \exp[v\psi(p)]\psi'^2(p)}{\int_0^1 \mathrm{d}p \exp[v\psi(p)]}\right)^{(-1)} . \tag{S23}$$

If we take

$$\psi(u) = \sin(2\pi u) \ , \tag{S24}$$

we find

$$\frac{D^*(\kappa)}{D^*(0)} = \left(1 - \frac{4\pi^2 \kappa k_\mathrm{B} T \gamma_{12}^2}{\lambda^2 \gamma_{11}^2} \frac{v I_1(v)}{I_0(v)}\right)^{(-1)} , \tag{S25}$$

where $I_n(z)$ denotes the modified Bessel function of the first kind.

This can be written as

$$\frac{D^*(\kappa)}{D^*(0)} \simeq (1 + \alpha) \ , \tag{S26}$$

where

$$\alpha = \frac{4\pi^2 \kappa k_\mathrm{B} T \gamma_{12}^2}{\lambda^2 \gamma_{11}^2} \frac{v I_1(v)}{I_0(v)} \ . \tag{S27}$$

In the next section, we will see that it is natural to write

$$\alpha = 4\pi^2 \epsilon \frac{\gamma_{12}^2}{\gamma_{11}^2} \frac{v I_1(v)}{I_0(v)} \ , \tag{S28}$$

where

$$\epsilon = \frac{\kappa k_\mathrm{B} T}{\lambda^2} \ , \tag{S29}$$

is the effective dimensionless compliance parameter that must be small for the above analysis to be valid.

The enhancement of the relative diffusion constant is clearly maximized by choosing small values of $\lambda$ so that the variable $q_2$ can influence the jumping rate between neighboring minima rather than being averaged out if $\lambda$ were large. We also see that the relative change depends on the function of the energy scale of the periodic potential

$$g(v) = \frac{v I_1(v)}{I_0(v)} \ , \tag{S30}$$

which is a monotonically increasing function of $v$. This means that the relative change is increased with the energy scale $\Delta U$ of the trap.

The Fokker-Planck equation in Eq. (S16) corresponds to the Ito stochastic differential equation

$$\frac{\mathrm{d}q_1}{\mathrm{d}t} = -\mu_\mathrm{e}(q_1)\phi'(q_1) + T\mu_\mathrm{e}'(q_1) + \sqrt{2T\mu_\mathrm{e}(q_1)}\eta(t) \ , \tag{S31}$$

where $\eta(t)$ is a zero-mean white noise with correlation function $\langle \eta(t)\eta(t')\rangle = \delta(t - t')$.

In the next section, we will see that this process has exactly the same diffusion constant as the process $q_1$ in the original hydrodynamically-coupled equations (S3,S4). Clearly, Eq. (S31) represents a temporal coarse graining of the original process $q_1$ on time scales greater than $\tau_2$. Finally, at very short time scales $t < \tau_2$, the microscopic time diffusion constant can similarly be deduced from Eq. (S3) and is given by

$$D_\mathrm{m} = k_\mathrm{B} T \mu_{11} = \frac{k_\mathrm{B} T \gamma_{22}}{\gamma_{11}\gamma_{22} - \gamma_{12}^2} \ . \tag{S32}$$

The expression $D_\mathrm{m}$ is simply the diffusion constant for the tracer in the absence of any hydrodynamic interactions with $q_2$ and no periodic potential; or, said otherwise, the bulk diffusivity of $q_1$.

## III. SUPPLEMENTARY NOTE 3: CORRECTION TO THE DIFFUSION BASED ON EXACT FORMULA FOR DIFFUSIVITY AND HOMOGENISATION THEORY

For small $\kappa$, $q_2$ is dominated by its fluctuations which scale *via* the equipartition of energy as

$$\delta q_2 \sim \sqrt{\kappa k_{\rm B} T} \ . \tag{S33}$$

Therefore, from now on, we will use the dimensionless position variables

$$q_1 = \lambda q \text{ and } q_2 = \sqrt{\kappa k_{\rm B} T} u \ , \tag{S34}$$

and again we also write $\beta\phi(q_1) = v\psi\left(\frac{q_1}{\lambda}\right)$. The coupled Langevin equations can be rewritten in terms of the dimensionless time

$$s = \frac{\mu_{11} k_{\rm B} T}{\lambda^2} t \tag{S35}$$

to yield

$$\frac{{\rm d}q}{{\rm d}s} = -v\psi'(q) - \frac{\mu_{12}}{\mu_{11}\sqrt{\epsilon}} u + \eta_1'(s) \tag{S36}$$

$$\frac{{\rm d}u}{{\rm d}s} = -\frac{\mu_{22}}{\mu_{11}}\frac{u}{\epsilon} - v\frac{\mu_{12}}{\mu_{11}\sqrt{\epsilon}}\psi'(q) + \frac{1}{\sqrt{\epsilon}}\eta_2'(s) \ , \tag{S37}$$

where $\eta_1'(s)$ and $\eta_2'(s)$ are Gaussian white noises with correlation functions

$$\langle \eta_i'(s)\eta_j'(s')\rangle = 2\frac{\mu_{ij}}{\mu_{11}}\delta(s-s') \ . \tag{S38}$$

and $\epsilon$ is defined in Eq. (S29).

The late time diffusion coefficient for $q_1$ is then given by via

$$2D^* t \simeq \langle q_1^2(t)\rangle = \lambda^2\langle q^2(s)\rangle \simeq 2\lambda^2 D_{q_1}^* s = 2D_q \mu_{11} k_{\rm B} T t \ . \tag{S39}$$

This means that the late time diffusion constant for $q_1$, $D^*$, is related to the late time diffusion constant of the scaled process $q$ by

$$D^* = \mu_{11} k_{\rm B} T D_q^*\left(\epsilon, v, \frac{\mu_{22}}{\mu_{11}}, \frac{\mu_{12}}{\mu_{11}}\right) \ , \tag{S40}$$

where we have given the explicit functional dependencies for $D_q^*$. We now carry out a multiscale analysis along the lines of [1, 2] where the correction of diffusion of a Brownian particle in a periodic potential due to inertial effects was given. Here, we will use the Stokes-Einstein relation to compute the diffusion constant and add an additional force $h$ acting on the variable $q$ in addition to the periodic potential. The Fokker-Planck equation for the variables $q$ and $u$ is then given by

$$\frac{\partial P}{\partial s} = HP \ , \tag{S41}$$

where $H$ is

$$HP = \left(H_0 + \frac{1}{\sqrt{\epsilon}}H_1 + \frac{1}{\epsilon}H_2\right)P \ , \tag{S42}$$

and the corresponding terms are given by

$$H_0 P = \frac{\partial}{\partial q}\left(\frac{\partial P}{\partial q} + [v\psi'(q) - h]P\right) \tag{S43}$$

$$H_1 P = \frac{\mu_{12}}{\mu_{11}}\left[u\frac{\partial P}{\partial q} + 2\frac{\partial^2 P}{\partial q \partial u} + (v\psi_1'(q) - h)\frac{\partial P}{\partial u}\right] \tag{S44}$$

$$H_2 P = \frac{\mu_{22}}{\mu_{11}}\frac{\partial}{\partial u}\left(\frac{\partial P}{\partial u} + uP\right) \ . \tag{S45}$$

The expansion we will make is based on small $\epsilon$ but also depends on the parameters $\mu_{22}/\mu_{11}$ and $\mu_{12}/\mu_{11}$, the separation of scales needed to implement the expansion is more precisely states as

$$1 \ll \frac{1}{\sqrt{\epsilon}}\frac{\mu_{12}}{\mu_{11}} \ll \frac{1}{\epsilon}\frac{\mu_{22}}{\mu_{11}} \ , \tag{S46}$$

and notably the second inequality implies that

$$\frac{\mu_{12}}{\mu_{11}} \ll \frac{1}{\sqrt{\epsilon}}\frac{\mu_{22}}{\mu_{11}} \ . \tag{S47}$$

It will be useful later to employ the adjoint $H^\dagger$ of $H$ given by

$$H^\dagger = H_0^\dagger + \frac{1}{\sqrt{\epsilon}}H_1^\dagger + \frac{1}{\epsilon}H_2^\dagger \ , \tag{S48}$$

where

$$H_0^\dagger f \ = \ \frac{\partial^2 f}{\partial q^2} - (v\psi'(q) - h)\frac{\partial f}{\partial q} \ , \tag{S49}$$

$$H_1^\dagger f \ = \ \frac{\mu_{12}}{\mu_{11}}\left(-u\frac{\partial f}{\partial q} + 2\frac{\partial^2 f}{\partial q \partial u} - [v\psi'(q) - h]\frac{\partial f}{\partial u}\right) \ , \tag{S50}$$

$$H_2^\dagger f \ = \ \frac{\mu_{22}}{\mu_{11}}\left(\frac{\partial^2 f}{\partial u^2} - u\frac{\partial f}{\partial u}\right) \ . \tag{S51}$$

We proceed by computing the average velocity (denoted here by $V$) of $q$ due to the applied field $h$ for a tracer particle which is started with the local equilibrium distribution within a single period of the potential. This is given by

$$V = \langle\frac{\mathrm{d}q}{\mathrm{d}s}\rangle = \int_{-\infty}^{\infty}\mathrm{d}q\,\mathrm{d}u\,\mathrm{d}u_0\int_\Lambda \mathrm{d}q_0\ q\frac{\mathrm{d}P(q,u|q_0,u_0;s)}{\mathrm{d}t}P_{\rm peq}(q_0,u_0) \ , \tag{S52}$$

where $\Lambda$ the integration range over $q_0$ is over a single period of length 1 and $P_{\rm peq}(q,u)$ is the equilibrium distribution for the variable $u$ and the periodized variable $q^* = q \bmod(1)$. Using the Fokker-Planck equation (S41) then gives

$$V = \int_{-\infty}^{\infty}\mathrm{d}q\,\mathrm{d}u\,\mathrm{d}u_0\int_\Lambda \mathrm{d}q_0 q\left[HP(q,u|q_0,u_0;s)\right]P_{\rm eq}(q_0,u_0) = \int_{-\infty}^{\infty}\mathrm{d}q\,\mathrm{d}u\,\mathrm{d}u_0\int_\Lambda \mathrm{d}q_0[H^\dagger q]P(q,u|q_0,u_0;s)P_{\rm peq}(q_0,u_0) \ , \tag{S53}$$

where

$$H^\dagger q = -v\psi'(q) + h - \frac{1}{\sqrt{\epsilon}}\frac{\mu_{12}}{\mu_{11}}u \ . \tag{S54}$$

We notice that $H^\dagger q$ is periodic in $q$, so we can write

$$\int_{-\infty}^{\infty}\mathrm{d}u\,\mathrm{d}u_0\,\mathrm{d}q\int_\Lambda \mathrm{d}q_0 P(q,u|q_0,u_0;s)P_{\rm peq}(q_0,u_0) = \int_{-\infty}^{\infty}\mathrm{d}u\,\mathrm{d}u_0\int_\Lambda \mathrm{d}q\,\mathrm{d}q_0[H^\dagger q]\sum_{n=-\infty}^{\infty}P(q+n,u|q_0,u_0;s)P_{\rm peq}(q_0,u_0) \ . \tag{S55}$$

However, $\sum_{n=-\infty}^{\infty}P(q+n,u|q_0,u_0;s) = P_p(q,u|q_0,u_0;s)$ is simply the transition density for the periodic variable $q^*$ on $\Lambda$. Clearly $P_p(q,u|q_0,u_0;s)$ satisfies the same Fokker-Planck equation as $P$ but restricted to $q \in \Lambda$ and with periodic boundary conditions.

As we have assumed that $q^*$ starts in equilibrium we find have

$$\int_{-\infty}^{\infty}\mathrm{d}u_0\int_\Lambda \mathrm{d}q_0 P_p(q,u|q_0,u_0;s)P_{\rm peq}(q_0,u_0) = P_{\rm peq}(q_0,u_0) \ , \tag{S56}$$

and we thus obtain the Stratonovich formula

$$V = \int_{-\infty}^{\infty}\mathrm{d}u\int_\Lambda \mathrm{d}q[H^\dagger q]P_{\rm peq}(q,u) \ , \tag{S57}$$

where

$$HP_{\rm peq}(q,u) = 0 \ , \tag{S58}$$

and has the periodic boundary condition $P_{\rm peq}(q+1,u) = P_{\rm peq}(q,u)$ . The Stokes-Einstein formula then gives the diffusion constant for the variable $q$ as

$$D_q^* = \frac{\partial V}{\partial h} = \int_{-\infty}^{\infty} \mathrm{d}u \int_\Lambda \mathrm{d}q \, \left[\frac{\partial H^\dagger}{\partial h} q\right] P_{\rm peq0}(q,u) + [H^\dagger q]\frac{\partial}{\partial h} P_{peq}(q,u) \ . \tag{S59}$$

In Eq. (S59), everything is evaluated at $h=0$, and the integral over $q$ is on a single period $\Lambda$ of the periodic potential $\psi$. We now denote

$$\frac{\partial}{\partial h} P_{\rm peq}(q,u) = r(q,u) \ , \tag{S60}$$

which obeys

$$Hr(q,u) - \frac{\partial}{\partial q} P_{\rm peq0}(q,u) - \frac{\mu_{12}}{\mu_{11}}\frac{1}{\sqrt{\epsilon}}\frac{\partial}{\partial u} P_{\rm peq0}(q,u) = 0 \ , \tag{S61}$$

and must in addition obey the integral constraint, due to the conservation of probability,

$$\int \mathrm{d}q \, \mathrm{d}u \frac{\partial}{\partial h} P_{\rm peq}(q,u) = \int \mathrm{d}q \, \mathrm{d}u \; r(q,u) = 0 \ . \tag{S62}$$

Now we note that Eq. (S54) gives

$$\frac{\partial H^\dagger}{\partial h} q = 1 \ , \tag{S63}$$

and consequently

$$D_q^* = \int_{-\infty}^{\infty} \mathrm{d}u \int_\Lambda \mathrm{d}q \; P_{\rm peq0}(q,u) + [H^\dagger q] r(q,u) \ . \tag{S64}$$

This simplifies to

$$D_q^* = 1 - \int \mathrm{d}q \, \mathrm{d}u \left( v\psi'(q) + \frac{1}{\sqrt{\epsilon}}\frac{\mu_{12}}{\mu_{11}} u \right) r(q,u) \ . \tag{S65}$$

The above expression can also be derived using established results on the theory of transport processes [4], or using a general Kubo-type formalism [5, 6]. In principle, the partial differential equation Eq. (S61) can be numerically solved and the integral in Eq. (S65) can be evaluated using the solution to give a numerically exact result for $D^*$ for all values of $\epsilon$, and we will show how this can be done in section (V). Here we compute the corrections due to finite $\kappa$ perturbatively for small $\epsilon$ while assuming that the variables $\mu_{ij}$ are of the same order to validate the expansion we make. We work in the adjoint representation writing

$$r(q,u) = P_{\rm peq0}(q,u) w(q,u) \ , \tag{S66}$$

which gives

$$H^\dagger w(q,u) + v\psi'(q) + \frac{u\mu_{12}}{\mu_{11}\sqrt{\epsilon}} = 0 \ , \tag{S67}$$

along with the integral constraint

$$\int_{-\infty}^{\infty} \mathrm{d}u \int_\Lambda \mathrm{d}q P_{\rm peq0}(q,u) w(q,u) = 0 \ . \tag{S68}$$

Setting $h=0$ in Eq. (S54), we see that Eq. (S67) has a solution $w(q,u) = q + A$, where $A$ is a constant. The integral constraint means that we must choose

$$A = -\langle q \rangle_{\rm eq} = \int_{-\infty}^{\infty} \mathrm{d}u \int_\Lambda \mathrm{d}q \; q P_{\rm peq0}(q,u) \ . \tag{S69}$$

However, we must ensure periodicity in $q$. We thus write

$$w(q,u) = q - \langle q\rangle_{\rm eq} + s(q,u) \ , \tag{S70}$$

where $s(q,u)$ has the boundary condition

$$1 + s(1,u) = s(0,u) \ , \tag{S71}$$

and obeys the equation

$$H^\dagger s(q,u) = 0 \ . \tag{S72}$$

Plugging this into the formula for $D_q^*$, we find

$$D_q^* = 1 - \int_{-\infty}^{\infty} {\rm d}u \int_\Lambda {\rm d}q P_{\rm peq0}(q,u)\left(v\psi'(q) + \frac{1}{\sqrt{\epsilon}}\frac{\mu_{12}}{\mu_{11}}u\right)(q - \langle q\rangle_{\rm eq} + s(q,u)) \ . \tag{S73}$$

The shift due to the term $\langle q\rangle_{\rm eq}$ gives zero contribution to the integral. The part of the integrand containing $q$ can be integrated by parts (the second term giving zero) to give

$$D_q^* = \frac{1}{Z_-}\exp\left[-v\psi(1)\right] - \int {\rm d}q\,{\rm d}u P_{\rm peq0}(q,u)\left(v\psi'(q) + \frac{1}{\sqrt{\epsilon}}\frac{\mu_{12}}{\mu_{11}}u\right)s(q,u) \ , \tag{S74}$$

and here we have explicitly taken $\Lambda = [0,1]$. In what follows, we denote

$$Z_\pm = \int_0^1 {\rm d}q \ \exp(\pm v\psi(q)). \tag{S75}$$

The first term in the integral on the right-hand side of Eq. (S74) can be integrated by parts to give the computationally useful formula

$$D_q^* = -\int_{-\infty}^{\infty} {\rm d}u \int_\Lambda {\rm d}q\, P_{\rm peq0}(q,u)\left(\frac{\partial s(q,u)}{\partial q} + \frac{1}{\sqrt{\epsilon}}\frac{\mu_{12}}{\mu_{11}}us(q,u)\right) \ . \tag{S76}$$

Now, we look for a solution of the form

$$s(q,u) = s_0(q,u) + \sqrt{\epsilon}s_1(q,u) + \epsilon s_2(q,u) + \sqrt{\epsilon}^3 s_3(q,u) + O(\epsilon^2) \ . \tag{S77}$$

This yields, expanding to order $\epsilon$,

$$\begin{aligned} H_2^\dagger s_0 &= 0 && \text{(S78)}\\ H_1^\dagger s_0 + H_2^\dagger s_1 &= 0 && \text{(S79)}\\ H_0^\dagger s_0 + H_1^\dagger s_1 + H_2^\dagger s_2 &= 0 && \text{(S80)}\\ H_0^\dagger s_1 + H_1^\dagger s_2 + H_2^\dagger s_3 &= 0 && \text{(S81)}\\ H_0^\dagger s_2 + H_1^\dagger s_3 + H_2^\dagger s_4 &= 0 \ . && \text{(S82)} \end{aligned}$$

The resulting diffusion constant is then given by

$$\begin{aligned} D_q^* &= -\frac{1}{\sqrt{\epsilon}}\int {\rm d}q\,{\rm d}u\,P_{\rm peq0}(q,u)\frac{\mu_{12}}{\mu_{11}}us_0(q,u) - \int {\rm d}q\,{\rm d}u\,P_{\rm peq0}(q,u)\left[\frac{\partial s_0(q,u)}{\partial q} + \frac{\mu_{12}}{\mu_{11}}us_1(q,u)\right]\\ &- \sqrt{\epsilon}\int {\rm d}q\,{\rm d}u\,P_{\rm peq0}(q,u)\left[\frac{\partial s_1(q,u)}{\partial q} + \frac{\mu_{12}}{\mu_{11}}us_2(q,u)\right]\\ &- \epsilon\int {\rm d}q\,{\rm d}u\,P_{\rm peq0}(q,u)\left[\frac{\partial s_2(q,u)}{\partial q} + \frac{\mu_{12}}{\mu_{11}}us_3(q,u)\right] \ , \end{aligned} \tag{S83}$$

and for notational simplicity, we have not explicitly written the integration ranges which are $u \in (-\infty,\infty)$ and $q \in [0,1]$. The equation (S79), $H_0^\dagger s_0(q,u) = 0$, shows that $s_0(q,u) = s_0(q)$ is independent of $u$. Note that in terms of the parity with respect to $u$, the operators $H_0^\dagger$ and $H_2^\dagger$ are even and $H_1^\dagger$ is odd. From the structure of the equations, this means that $s_n(q,u)$ is even in $u$ for $n$ even and odd in $u$ for $n$ odd. As a consequence we find that

$$D_q^* = -\int \mathrm{d}q\, \mathrm{d}u\, P_{\mathrm{peq0}}(q,u)\left[\frac{\partial s_0(q,u)}{\partial q} + \frac{\mu_{12}}{\mu_{11}} u s_1(q,u)\right] - \epsilon \int \mathrm{d}q\, \mathrm{d}u\, P_{\mathrm{peq0}}(q,u)\left[\frac{\partial s_2(q,u)}{\partial q} + \frac{\mu_{12}}{\mu_{11}} u s_3(q,u)\right] \ , \tag{S84}$$

and it is easy to see that the higher order corrections are also integer powers of $\epsilon$. We now consider Eq. (S79): $H_1^\dagger s_0 + H_2^\dagger s_1 = 0$. From this we see that solution for $s_1$ exists if the Fredholm alternative condition holds (the solvability condition)

$$\int \mathrm{d}u P_{\mathrm{eq}}(u) H_1^\dagger s_0(q) = 0 \ , \tag{S85}$$

where $P_{\mathrm{eq}}(u) = \frac{1}{\sqrt{2\pi}}\exp\left(-\frac{u^2}{2}\right)$, the marginal equilibrium probability distribution of $u$, is in the kernel of the adjoint of $H_2$, that is to say $H_2^\dagger$. We find that

$$H_1^\dagger s_0(q) = -\frac{\mu_{12}}{\mu_{11}} u \frac{\partial s_0(q)}{\partial q} \ , \tag{S86}$$

and this gives

$$s_1(q,u) = -\frac{\mu_{12}}{\mu_{22}} u \frac{\partial s_0(q)}{\partial q} \ . \tag{S87}$$

Now, consider the equation Eq. (S80), $H_0^\dagger s_0 + H_1^\dagger s_1 + H_2^\dagger s_2 = 0$. Here, we find that

$$H_1^\dagger s_1 = \frac{\mu_{12}^2}{\mu_{11}\mu_{22}}\left([u^2-2]\frac{\partial^2 s_0(q)}{\partial q^2} + v\psi'(q)\frac{\partial s_0(q)}{\partial q}\right) \ . \tag{S88}$$

Integrating Eq. (S80) with respect to $P_{\mathrm{eq}}(u)$ then gives the solvability condition for $s_2$ as

$$H_0^\dagger s_0 + \frac{\mu_{12}^2}{\mu_{11}\mu_{22}}\left(-\frac{\partial^2 s_0(q)}{\partial q^2} + v\psi'(q)\frac{\partial s_0(q)}{\partial q}\right) = 0 \ , \tag{S89}$$

which is simply

$$\left(1 - \frac{\mu_{12}^2}{\mu_{11}\mu_{22}}\right) H_0^\dagger s_0 = 0 \ . \tag{S90}$$

This has the solution

$$s_0(q) = A\int_0^q \exp[v\psi(p)]\,\mathrm{d}p \ , \tag{S91}$$

where $A$ is a constant. Note that we can choose $s_0(0) = 0$ without loss of generality as $D_q^*$ is independent of any constant term in $s(q)$. Using the boundary condition $s_0(1) = -1$ then gives

$$s_0(q) = -\frac{\int_0^q \exp[v\psi(p)]\,\mathrm{d}p}{Z_+} \ , \tag{S92}$$

and so

$$s_0'(q) = -\frac{\exp[v\psi(q)]}{Z_+} \ . \tag{S93}$$

This then yields

$$s_1(q,u) = \frac{\mu_{12}}{\mu_{22}} u \frac{\exp[v\psi(q)]}{Z_+} \ . \tag{S94}$$

Importantly, we see that $s_1(q,u)$ is periodic in $q$ and so the solution for $s$ at order $\sqrt{\epsilon}$ obeys the full boundary condition *via* the term $s_0(q)$ solely, all the other corrections must consequently be periodic. This gives the $O(0)$ contribution to $D_q^*$

$$D_q^*(0) = \frac{1}{Z_+ Z_-}\left[1 - \frac{\mu_{12}^2}{\mu_{22}\mu_{11}}\right] . \tag{S95}$$

Returning to Eq. (S80) $H_0^\dagger s_0 + H_1^\dagger s_1 + H_2^\dagger s_2 = 0$ we find that

$$\frac{\mu_{12}^2}{\mu_{11}\mu_{22}}[u^2-1]\frac{\partial^2 s_0(q)}{\partial q^2} + H_2^\dagger s_2 = 0 . \tag{S96}$$

This satisfies the solvability condition, and has solution

$$s_2(q,u) = u^2\frac{\mu_{12}^2}{2\mu_{22}^2}\frac{\partial^2 s_0(q)}{\partial q^2} + s_{20}(q) , \tag{S97}$$

where $s_{20}(q)$ is undetermined (but must be periodic). Now solving Eq. (S81), $H_0^\dagger s_1 + H_1^\dagger s_2 + H_2^\dagger s_3$, we find the periodic solution

$$s_3(q,u) = -u\frac{\mu_{12}}{\mu_{22}}s_{20}'(q) + \frac{\mu_{12}u\exp[v\psi(q)]\left(\mu_{12}^2u^2v^2\psi'(q)^2 + v\psi''(q)[6(\mu_{11}\mu_{22}-\mu_{12}^2) - u^2\mu_{12}^2]\right)}{6Z_+\mu_{22}^3} . \tag{S98}$$

To determine $s_{20}(q)$, we use the Fredholm alternative for solvability for Eq. (S82), $H_0^\dagger s_2 + H_1^\dagger s_3 + H_0^\dagger s_4 = 0$, which yields after some lengthy algebra

$$s_{20}(q) = \int_0^q \exp[v\psi(p)]\,\mathrm{d}p\left[A' + \frac{\mu_{12}^2(v^2\psi'(p)^2 - v\psi''(p))}{2\mu_{22}^2 Z_+}\right] , \tag{S99}$$

where we have again taken $s_{20}(0) = 0$, and so the (periodic) boundary condition at $q = 1$ is $s_{20}(1) = 0$ (recalling that the inhomogeneous boundary condition is satisfied by $s_0(q)$), we thus have

$$A'Z_+ + \int_0^1 \exp[v\psi(p)]\,\mathrm{d}p\,\frac{\mu_{12}^2(v^2\psi'(p)^2 - v\psi''(p))}{2\mu_{22}^2 Z_+} = 0 , \tag{S100}$$

and so

$$A' = -\frac{\mu_{12}^2}{\mu_{22}^2 Z_+^2}\int_0^1 \exp[v\psi(p)]v^2\psi'(p)^2\mathrm{d}p . \tag{S101}$$

This then yields

$$s_{20}'(q) = \left[-\frac{\mu_{12}^2}{\mu_{22}^2 Z_+^2}\int_0^1 \exp[v\psi(p)]v^2\psi'(p)^2\mathrm{d}p + \frac{\mu_{12}^2(v^2\psi'(p)^2 - v\psi''(p))}{2\mu_{22}^2 Z_+}\right]\exp[v\psi(q)] . \tag{S102}$$

From the above, we find that the correction to $D_q^*$ to $O(\epsilon)$ is

$$\Delta D_{q2}^* = \epsilon v^2\frac{(\mu_{11}\mu_{22}-\mu_{12}^2)}{\mu_{22}\mu_{11}}\frac{\mu_{12}^2}{\mu_{22}^2}\frac{\int_0^1\exp[v\psi(p)]\psi'(p)^2\mathrm{d}p}{Z_- Z_+^2} . \tag{S103}$$

Using this to compute $D^*$ then gives the formula derived from the heuristic argument presented in section (II).

## IV. SUPPLEMENTARY NOTE 4: THE COMPLIANT LIMIT $\epsilon \gg 1$

In the limit $\epsilon \gg 1$, one can carry out a similar analysis to the limit $\epsilon \ll 1$. The Kubo-type formulae derived in the previous section still hold and the equation for $s(q,u)$ is now written as

$$\left[\epsilon H_0 + \sqrt{\epsilon}H_1 + H_2\right]s(q,u) , \tag{S104}$$

and the formula for $D_q^*$ is unchanged.

Here, we find that one need to write a perturbative expansion of the form

$$s(q,u) = \sqrt{\epsilon} s_0(q,u) + s_1(q,u) + \frac{1}{\sqrt{\epsilon}} s_2(q,u) + O(\frac{1}{\epsilon}) \ . \tag{S105}$$

We find that

$$\begin{aligned} H_0^\dagger s_0 &= 0 && \text{(S106)} \\ H_0^\dagger s_1 + H_1^\dagger s_0 &= 0 && \text{(S107)} \\ H_0^\dagger s_2 + H_1^\dagger s_1 + H_2^\dagger s_0 &= 0 \ . && \text{(S108)} \end{aligned}$$

The equation $H_0 s_0 = 0$ then gives $s_0(q,u) = s_0(u)$, and so, to leading order, we find

$$D_q^* = -\int_{-\infty}^{\infty} \mathrm{d}u \int_\Lambda \mathrm{d}q\, P_{\mathrm{peq}0}(q,u) \left( \frac{\partial s_1(q,u)}{\partial q} + \frac{\mu_{12}}{\mu_{11}} s_0'(u) \right) + O(\frac{1}{\sqrt{\epsilon}}) \ . \tag{S109}$$

We impose the non-periodic boundary condition on the $O(1)$ term $s_1(q,u)$

$$s_1(0,u) = 0 \ ; \ 1 + s_1(1,u) = 0 \ . \tag{S110}$$

The term $s_1$ obeys the equation $H_1 s_0 + H_0 s_1 = 0$, where $H_1 s_0 = -\frac{\mu_{12}}{\mu_{11}} v\psi'(q) s'(u)$. We notice that

$$\int_\Lambda \mathrm{d}q\, P_{\mathrm{eq}}(q) H_1 s_0 = 0 \ , \tag{S111}$$

where

$$P_{\mathrm{eq}}(q) = \frac{\exp[-v\psi(q)]}{Z_-} \ , \tag{S112}$$

and so this equation obeys the solvability criterion. The solution is given by

$$s_1(q,u) = -q\frac{\mu_{12} s_0'(u)}{\mu_{11}} + \frac{1}{Z_+}\left[\frac{\mu_{12} s_0'(u)}{\mu_{11}} - 1\right] \int_0^q \mathrm{d}p \ \exp[v\psi(p)] \ . \tag{S113}$$

Using this result in Eq. (S109) then gives to leading order

$$D_q^* = \frac{1}{Z_+ Z_-}\left[1 - \frac{1}{\sqrt{2\pi}} \frac{\mu_{12}}{\mu_{11}} \int_{-\infty}^{\infty} \mathrm{d}u \exp(-\frac{u^2}{2}) s_0'(u)\right] + O(\frac{1}{\sqrt{\epsilon}}) \ . \tag{S114}$$

Finally, the equation for $s_0(u)$ can be found by using the solvability condition for $s_2$

$$0 = \int_\Lambda \mathrm{d}q \ P_{\mathrm{eq}}(q) \left[H_0^\dagger s_2 + H_1^\dagger s_1 + H_2^\dagger s_0\right] = H_2^\dagger s_0(u) + \int_\Lambda \mathrm{d}q \ P_{\mathrm{eq}}(q) H_1^\dagger s_1(q,u) \ . \tag{S115}$$

After some algebra, the solution which is integrable with respect to the equilibrium measure $P_{\mathrm{eq}}(u)$ for $u$ is

$$s_0(u) = \frac{u\mu_{12}\mu_{11}}{(\mu_{11}\mu_{22} - \mu_{12}^2) Z_+ Z_- + \mu_{12}^2} \ . \tag{S116}$$

Using this in Eq. (S114) then gives

$$D_q^* = \frac{\mu_{11}\mu_{22} - \mu_{12}^2}{Z_+ Z_- (\mu_{11}\mu_{22} - \mu_{12}^2) + \mu_{12}^2} + O(\frac{1}{\sqrt{\epsilon}}) \ . \tag{S117}$$

Note that when there is no periodic potential ($v = 0$) one has $Z_+ = Z_- = 1$, and we find

$$D_q^*(\epsilon = \infty) = k_{\mathrm{B}} T \frac{\mu_{11}\mu_{22} - \mu_{12}^2}{\mu_{22}} = \frac{k_B T}{\gamma_{11}} \tag{S118}$$

which is the same result for the diffusion constant in the non-compliant case $\epsilon = 0$ and with $v = 0$. In terms of the unscaled variables this then gives

$$D^* = k_{\mathrm{B}} T \mu_{11} \frac{\mu_{11}\mu_{22} - \mu_{12}^2}{\lambda^{-2} Z_+ Z_- (\mu_{11}\mu_{22} - \mu_{12}^2) + \mu_{12}^2} + O(\frac{1}{\sqrt{\epsilon}}) \ . \tag{S119}$$

## V. SUPPLEMENTARY NOTE 5: NUMERICAL COMPUTATION OF $D^*$ FROM THE KUBO FORMULA

The diffusion constant $D_q^*$ related to $D^*$ by Eq. (S40) can be numerically computed by solving Eq. (S61) with periodic boundary conditions and the integral constraint Eq. (S62). This can be carried out by using the FlexPDE Multiphysics Software For Partial Differential Equations. The numerical domain is finite and taken over $[-1/2, 1/2]$ for the variable $q$ with periodic boundary conditions. The variable $u$ is unbounded, but we take a finite domain with $u \in [-L, L]$ (typically, it suffices to take $L = 5$), and impose no-flux boundary conditions at the surfaces $u = \pm L$. This means that the effective potential for $u$ becomes infinite at the edges, but in practice the variable $r(q, u)$ has decayed to zero at these values of $u$. Examples are shown in Figs. 4, S1, and S2, for various parameters.

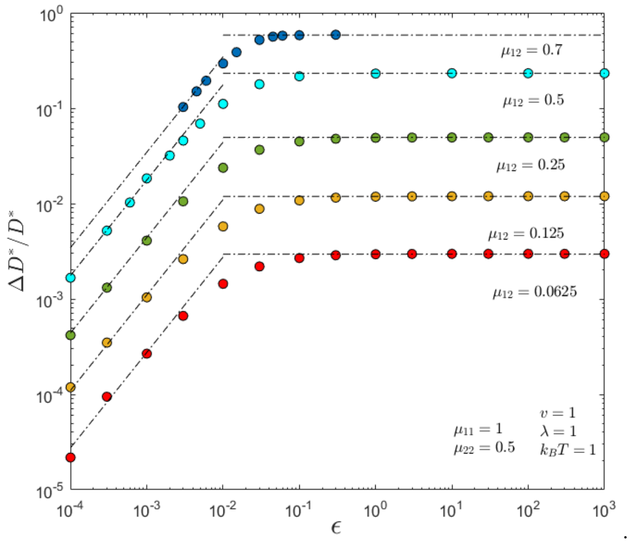


FIG. S1. Relative increase $\Delta D^*/D^* = [D^*(\epsilon) - D^*(0)]/D^*(0)$ of the late-time diffusion coefficient $D^*$ of the test particle as a function of the dimensionless compliance $\epsilon$, for $\mu_{11}$= 1.0, $\mu_{22} = 0.5$, $v = 1.0$, $\lambda = 1.0$, $k_\mathrm{B}T = 1.0$, and various $\mu_{12}$ as indicated. The symbols correspond to the Kubo formula of Eq. (S65). The dash-dotted lines correspond to the small-$\epsilon$ prediction of Eq. (5), and the large-$\epsilon$ prediction of Eq. (11).

## VI. SUPPLEMENTARY NOTE 6: HEURISTIC ARGUMENT FOR MODIFICATION OF DIFFUSION WITH INERTIA

Here, we show how the heuristic method proposed beforehand can be used to compute the effective diffusion constant for a particle due to first order inertial effects, recovering the result of [1, 2]. This calculation also demonstrates how a periodic potential can be applied to detect the signature of inertia on late-time diffusion.

We consider the zero-temperature equation

$$m\frac{\mathrm{d}v}{\mathrm{d}t} + \gamma v = -\phi'(q) \ , \tag{S120}$$

where $v = \frac{\mathrm{d}q}{\mathrm{d}t}$ is the particle's velocity.

In operator notation, this formally gives

$$v(t) = -\frac{1}{\gamma}\left(1 + \tau\frac{\mathrm{d}}{\mathrm{d}t}\right)^{-1}\phi'(q) \ , \tag{S121}$$

where $\tau = \frac{m}{\gamma}$ is the short timescale over which the velocity relaxes and which would be difficult to resolve experimentally.

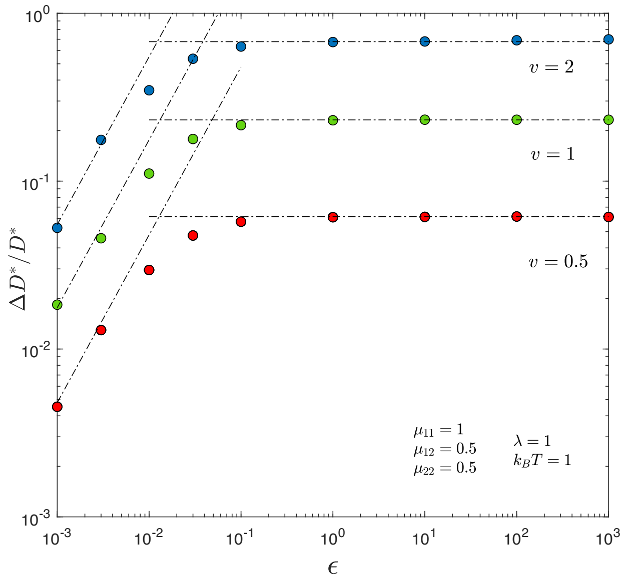


FIG. S2. Relative increase $\Delta D^*/D^* = [D^*(\epsilon) - D^*(0)]/D^*(0)$ of the late-time diffusion coefficient $D^*$ of the test particle as a function of the dimensionless compliance $\epsilon$, for $\mu_{11}$= 1.0, $\mu_{12} = 0.5$, $\mu_{22} = 0.5$, $\lambda = 1.0$, $k_\mathrm{B}T = 1.0$, and various $v$ as indicated. The symbols correspond to the Kubo formula of Eq. (S65). The dash-dotted lines correspond to the small-$\epsilon$ prediction of Eq. (5), and the large-$\epsilon$ prediction of Eq. (11).

Now, expanding to first order in $\tau$, we find

$$v(t) = -\frac{1}{\gamma}\phi'(q) + \frac{\tau}{\gamma}\phi''(q)v(t) \ , \tag{S122}$$

and so

$$\frac{\mathrm{d}q}{\mathrm{d}t} = v(t) = -\mu_\mathrm{e}(q)\phi'(q) \ , \tag{S123}$$

where

$$\mu_\mathrm{e}(q) = \frac{1}{\gamma\left[1 - \frac{\tau}{\gamma}\phi''(q)\right]} \ . \tag{S124}$$

The diffusion constant is thus

$$D_\mathrm{e}(q) = \frac{k_\mathrm{B}T}{\gamma\left[1 - \frac{\tau}{\gamma}\phi''(q)\right]} \ . \tag{S125}$$

Now, applying the Lifson-Jackson formula, we obtain

$$D^* = \frac{k_\mathrm{B}T\lambda^2}{\gamma\int_0^\lambda \mathrm{d}q\exp[\beta\phi(q)]\left[1 - \frac{m}{\gamma^2}\phi''(q)\right]\int_0^\lambda \mathrm{d}q\exp[-\beta\phi(q)]} \ . \tag{S126}$$

This can be rewritten as

$$D^* = \frac{k_\mathrm{B}T\lambda^2}{\gamma\int_0^\lambda \mathrm{d}q\exp[\beta\phi(q)]\left[1 + \frac{m}{k_\mathrm{B}T\gamma^2}\phi'(q)^2\right]\int_0^\lambda \mathrm{d}q\exp[-\beta\phi(q)]} \ , \tag{S127}$$

or to the same order

$$D^* = \frac{k_\mathrm{B}T\lambda^2}{\gamma\int_0^\lambda \mathrm{d}q\exp[\beta\phi(q)]\int_0^\lambda \mathrm{d}q\exp[-\beta\phi(q)]}\left(1 - \frac{m}{k_\mathrm{B}T\gamma^2}\frac{\int_0^\lambda \mathrm{d}q\exp[\beta\phi(q)]\phi'(q)^2}{\int_0^\lambda \mathrm{d}q\exp[\beta\phi(q)]}\right) \ , \tag{S128}$$

which recovers the result given in [1, 2] via multiscale analysis. Notice that in this particular case, the effect of inertia is to reduce the late time diffusion constant, and again, it is necessary to impose an external periodic potential to produce a measurable effect on the late time diffusivity.

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